

BEZAWADA HEMANTH CHOWDARY

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EDUCATION

B.E. Computer Science & Engineering Jul 2022 – Jun 2026

Saveetha Institute of Medical and Technical Sciences (SIMATS) · Chennai, India

- CGPA: 8.65 / 10 · Relevant coursework: Data Structures & Algorithms, Operating Systems, Machine Learning, Database Systems, Computer Networks, Distributed Computing

TECHNICAL SKILLS

Languages: Python (advanced), SQL, R; learning Go

Systems & Concurrency: asyncio, multi-threading (ThreadPoolExecutor), multiprocessing, Celery, Redis, task queues, REST APIs, WebSockets, FastAPI

AI / ML: PyTorch, TensorFlow, HuggingFace Transformers, Scikit-learn, NLP, Computer Vision, LLM integration

Data & Infra: Pandas, NumPy, PostgreSQL, Docker, Git, Linux, Google Cloud Platform, Microsoft Azure, Power BI

Algorithms: Graph traversal, Dynamic Programming, Binary Search, Sliding Window, Prefix Sums, Hash Maps, BFS/DFS

Certifications: Oracle Cloud Infrastructure DS Professional · Google Data Analytics · Google AI Essentials · Harvard CS50

EXPERIENCE

Software Engineering Intern — AI Data Systems Sep 2025 – Dec 2025

Lesoko · Chennai, India

- Engineered a Python-based automated validation pipeline processing 10,000+ NLP and CV annotation samples end-to-end; implemented rule-based and statistical quality checks achieving 98%+ accuracy, reducing human review overhead by 40%.
- Designed annotation schema versioning system across 3 concurrent ML projects alongside engineers, reducing iteration cycles by 30%, analogous to schema migration management in production data systems.
- Built edge-case detection logic using statistical outlier analysis to surface 500+ systematically mislabeled samples before model training, preventing downstream bias propagation in production AI pipelines.

PROJECTS

Distributed Web Crawler & Search Engine Aug – Oct 2026

Python · asyncio · ThreadPoolExecutor · Redis · FastAPI · SQLite · github.com/HemanthChowdaryBezawada/distributed-crawler

- Building a concurrent web crawler using Python asyncio and ThreadPoolExecutor that crawls 10,000+ pages simultaneously; implemented a Redis-backed distributed task queue so multiple workers claim URLs without duplication or race conditions.
- Designed a TF-IDF + BM25 indexing system over crawled content with full-text search via a FastAPI REST endpoint; achieved sub-100ms query latency on a 50,000-document index through inverted-index caching.
- Added politeness controls (per-domain rate limiting), robots.txt compliance, and fault-tolerant retry logic with exponential backoff, production-grade patterns for distributed crawling at scale.

Real-Time NLP Analysis Platform with Concurrent Processing Jun – Jul 2026

Python · FastAPI · Celery · Redis · HuggingFace · WebSocket · Docker · github.com/HemanthChowdaryBezawada/nlp-pipeline

- Building a real-time NLP platform where users submit text via WebSocket and background Celery workers run sentiment analysis, named entity recognition, and summarization concurrently across a 4-worker pool; reduced end-to-end latency by 62% vs sequential processing.
- Implemented priority task queue (urgent vs. batch jobs), request deduplication via Redis key hashing, and result caching with TTL, synchronization patterns that prevent thundering-herd failures under peak load.
- Containerized with Docker Compose (FastAPI + Celery + Redis + Flower monitoring); designed rate-limiting middleware handling 500+ concurrent WebSocket connections in load tests.

Parallelized Multilingual Data Processing Pipeline Aug – Nov 2024

Python · multiprocessing · HuggingFace Transformers · Pandas

- Designed and implemented a parallelized preprocessing pipeline for 50,000+ multilingual translation pairs using Python multiprocessing.Pool; achieved 4.1x speedup over sequential baseline by eliminating I/O bottlenecks with buffered chunk reads.
- Benchmarked BPE, WordPiece, and SentencePiece tokenization strategies across 4 language pairs; optimal 32K vocabulary configuration reduced OOV token rate from 12% to 4%, improving Seq2Seq model convergence by ~15%.

Customer Behavior Prediction — End-to-End ML System Jan – May 2024

Python · Scikit-learn · Pandas · Statistical Modeling · Best Poster Award, Tech Star Summit 2024

- Built a complete ML pipeline (ingestion → cleaning → feature engineering → model training → evaluation); chi-square feature selection across 15+ engineered features cut dimensionality by 45%, improving targeting precision by 25-35% over baseline.
- Implemented stratified 5-fold cross-validation with reproducible seeds and hyperparameter logging; results presented to 200+ industry attendees, won Best Poster Award at Tech Star Summit, May 2024.

AWARDS & LEADERSHIP

- Best Poster Award — Tech Star Summit 2024 · "Customer Response Analytics for Insurance Outreach" · Competitive review, 200+ industry attendees.
- Organized 4 technical workshops on Python, DSA, and ML for 1,000+ undergraduates; mentored 8 students, 3 of whom independently published research at university symposiums.